NAVYA BOGA

LinkedIn | 860-938-5156 | NavyaBoga.com | navyaboga1109@gmail.com | GitHub

Skills

- **Languages:** Java, Python, TypeScript, JavaScript, SQL, C++ **Backend & APIs**: Node.js, REST APIs, Microservices, GraphQL
- **Front-end:** React.js, TypeScript, JavaScript, Redux Toolkit (RTK Query), HTML, CSS, AWS Cloudscape Design System
- **Cloud & Infrastructure**: AWS, AWS CDK, CloudFormation, Docker, Kubernetes, Linux, CI/CD, Distributed Systems, CloudWatch
- **Testing & Tools**: Vitest, React Testing Library, Mockito, Jenkins, Git, Agile Development

Education

- Master of Science - Data Science — **The University of Memphis** Memphis, TN, USA (08/2023 - 05/2025)
- Bachelor of Engineering – Computer Science — **Osmania University** (05/2019 - 07/2023)

Experience

Software Development Engineer - Amazon Web Services, S3 Console **Jul 2025 – Present**

• Built and launched full-stack features for the **AWS S3 Console** used by millions of AWS customers across **React/TypeScript frontend systems, backend integrations, cloud infrastructure, and deployment tooling.**

• Architected a multi-step Storage Lens wizard for S3 Tables with reusable **Redux Toolkit state management**, validation workflows, and **API-driven resource orchestration**.

• Designed reusable **Container/Presenter architecture patterns** that improved component reuse, maintainability, and testability across multiple Storage Lens workflows.

• Developed multi-region resource management workflows with AWS Organizations integration, org-scoped access controls, and parallel cross-region data fetching using **RTK Query** and Promise.allSettled.

• Engineered real-time analytics dashboards with **KPI metrics, sortable trend tables, sparkline visualizations**, and a recommendation engine performing anomaly detection across 19 storage metrics.

• Built scalable API integration layers for **S3, EC2, S3 Tables, and AWS Organizations** services with resilient partial-failure handling and reusable connector abstractions.

• Improved deployment reliability by designing **AWS CDK wave-isolation logic** for sovereign cloud regions and extending monitoring/canary coverage across **Gamma and Production environments**.

• Resolved production issues involving Redux state synchronization, deployment pipeline failures, and infrastructure reliability through debugging and architecture improvements.

• Developed comprehensive integration and infrastructure test suites using **Vitest, React Testing Library, and Mockito** while maintaining 99%+ test coverage across new feature development.

Technologies: React.js, TypeScript, Redux Toolkit, RTK Query, AWS CDK, CloudFormation, CloudWatch, S3, EC2, Java, Node.js, Vitest, Mockito

Graduate Assistant - Center for Information Assurance, The University of Memphis **December 2024 – May 2025**

• Developed and maintained web-based cybersecurity training modules and instructional platforms **using Python, JavaScript, HTML/CSS, and data visualization tools**, supporting coursework focused on network defense, risk management, and incident response.

• Built interactive dashboards and technical content delivery workflows for cybersecurity research and training under the National Cybersecurity Preparedness Consortium (NCPC) in collaboration with the Department of Homeland Security.

Data Engineer Intern - The University of Tennessee Health Science Center **May 2024 -August 2024**

• Performed large-scale genomic data analysis and preprocessing using **Python, R, pandas, and NumPy**, improving mutation detection accuracy across mitochondrial DNA datasets. Optimized data processing and visualization workflows using Jupyter Notebook, Matplotlib, Shell scripting, and Unix-based tooling. **Technologies:** Python, R, Pandas, NumPy, Unix, Shell Scripting

Java Ignite Trainee and Intern Larsen & Toubro Infotech Mindtree **Nov 2022 - July 2023**

• Developed Java-based cloud applications using AWS services including **EC2, S3, RDS, and Lambda** for scalable deployment workflows and backend integrations. Built and maintained CI/CD pipelines using Jenkins, Docker, Kubernetes, and Git, reducing deployment errors, improving operational efficiency.

• Worked on **REST API integrations, Agile development workflows**, and monitoring automation for distributed application environments.

Technologies: Java, AWS, Docker, Kubernetes, Jenkins, REST APIs, Git

Web Application Developer Intern - Suntek Corporations Private Limited **July 2022 - Aug 2022**

• Built a facial recognition attendance system using **Python, OpenCV, NumPy, and machine learning techniques including CNN, SVM, and HOG**, improving image processing performance and reducing false identification rates. (REPOSITORY)

Web Application Developer Intern - Defense Research and Development Organization **May 2022 - June 2022**

• Developed a real-time intranet chat application using **React.js, PHP, SQL, Python, WebSockets, and Django Channels**, supporting 100+ concurrent users with low-latency secure messaging. (REPOSITORY)

Projects

- **Forged Emblem Detection System:** Designed to identify counterfeit emblems with 97.5% accuracy using advanced image processing techniques (Transfer Learning) to ensure accurate detection and authentication of legitimate emblems. **Tech Stack**: Python, TensorFlow, PIL (Python Imaging Library), Streamlit, NumPy, SSL, network security and Infrastructure system design. (REPOSITORY)
- **E-Commerce Platform Development**: Engineered a full-fledged e-commerce system with secure user authentication, dynamic product listings, and seamless checkout and payment integration. Achieved optimized data handling and performance using MySQL and Neo4j for scalability. **Tech Stack:** React.js, Node.js, MySQL, Neo4j, Stripe API, PHP, SSL encryption, and RESTful API design. (REPOSITORY)

Courses and Certifications

- **ORACLE**: Cloud Infrastructure 2024 Generative AI Certified Professional
- **AWS**: Cloud Foundations